Component-Level Analysis of Web Interface Vulnerability Under Simulated CVD

Luke Vassallo
Institute of Information & Communication Technology
Malta College or Arts, Science & Technology
Corradino Hill
Paola PLA 9032
luke.vassallo.g52924@mcast.edu.mt
Github Link: https://github.com/LukeVassallo18/ResearchDesign2LVCCD

***Abstract*—This study evaluates the contrast accessibility of web interface components under simulated Colour Vision Deficiency (CVD) conditions. A Python-based pipeline was used to scan 24 public websites, classify UI components, simulate protanopia, deuteranopia, and tritanopia, and assess foreground–background contrast against WCAG 2.2 thresholds. The dataset included 22,950 components. Results showed stable overall contrast performance across all CVD conditions, with an 85% pass rate. However, component-level differences were identified, with alert and input components performing strongest, while links showed the highest vulnerability. The study highlights the value of component-level accessibility analysis and shows that WCAG contrast metrics provide a useful baseline, but may not fully capture CVD-related perceptual differences.**

***Index Terms*—Keywords go Here**

I. Introduction

Web accessibility ensures that websites and online systems align with the POUR principles: Perceivable, Operable, Understandable, and Robust, supporting users with visual, auditory, motor, and cognitive differences [1]. As the web increasingly functions as a gateway to education, employment, and services, inaccessible design could risk excluding a significant portion of the population [1], [2].

An estimated 300 million individuals worldwide experience some form of Colour Vision Deficiency (CVD), including around 1 in 12 men and 1 in 250 women [3]. Because colour is widely used to communicate states such as alerts, navigation cues, and data visualisations, users with CVD may experience reduced clarity when interacting with colour-dependent interfaces [3], [4].

While prior studies provide insight into colour perception and interface usability, many evaluate interfaces as a whole or focus on predefined colour schemes [4], [5]. Limited research has examined how individual User Interface (UI) component categories behave under simulated CVD conditions across multiple websites, revealing a gap in structured component-level contrast analysis [2], [6].

A. Positioning

As shown in Fig.1, the Research Onion represents the methodological decisions guiding this study. This research adopts a pragmatic approach, focusing on selecting methods that best address the research problem rather than adhering to a single philosophical viewpoint.

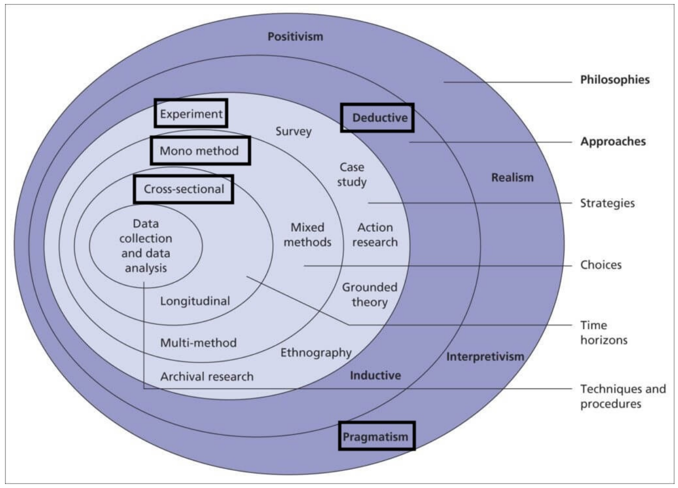

Fig. 1: Research onion illustrating philosophical and methodological positioning

The purpose of this study is to evaluate the individual accessibility of web interface components under simulated CVD conditions using a quantitative, component-level analysis. Specifically, the study examines how contrast ratios of UI components change when normal vision is compared to simulated protanopia, deuteranopia, and tritanopia conditions.

A deductive, mono-method quantitative approach is employed in the study, as outlined in [2]. The experimental design involves analysing websites as previously stated. The independent variable is the simulated vision condition, while the dependent variable is the contrast performance of UI components, measured against the Web Content Accessibility Guidelines (WCAG) 2.2 [1].

The study follows a cross-sectional time horizon, where data is collected and analysed at a single point in time across multiple websites.

B. Hypothesis

Simulated CVD conditions will reveal measurable contrast degradation across UI component categories.

C. Research Aim and Questions

This study aims to evaluate how different UI component categories perform in terms of foreground–background contrast under simulated CVD conditions [3], guided by the following questions:

- To what extent do UI components maintain acceptable contrast thresholds under simulated CVD conditions?
- How does contrast performance vary across component categories under simulated CVD conditions?
- Which UI components demonstrate the highest levels of contrast vulnerability across multiple websites?

II. Literature Review

A. Accessibility Evaluation Approaches

Web accessibility evaluation methods are split into several categories; automated tools, expert inspection, and user testing [6]. Automated tools efficiently detect vulnerabilities such as missing alternative text or insufficient contrast ratios by scanning actual values against WCAG standards [6]. However, due to limited contextual understanding, studies which used automated tools have shown that no single tool captures all vulnerabilities, with many producing inconsistent or incomplete reports [6], [2]. Expert inspection through heuristic evaluation and WCAG audits provide deeper qualitative analysis, with drawbacks in subjectivity and scalability [2]. User testing remains the most valid approach, capturing practical barriers that neither tools or experts reliably detect [6]. However, this method is resource-intensive and findings are difficult to generalise beyond the participant sample. Hybrid approaches that combine two or more of these methods improve overall coverage but inherit the cost and complexity of each [6].

B. Evaluation Metrics and Their Limitations

Recent accessibility studies commonly use WCAG-based contrast metrics, reporting pass rates and failure percentages to evaluate compliance at scale [7]. These approaches provide clear and reproducible measurements, identifying widespread contrast issues across websites.

However, results are typically aggregated at the page or site level, which masks how individual UI components contribute to accessibility failures [6]. Although effective in identifying the presence of issues, these metrics do not explain where failures occur within the interface.

This limitation motivates a component-level approach, where WCAG contrast metrics are applied to individual UI elements, enabling more precise identification of accessibility vulnerabilities.

C. CVD Simulation and Colour Contrast Research

Additionally, a separate study has addressed CVD through computational simulation and perceptual modelling. Simulation methods such as Daltonlens apply colour vision models to replicate visual impairments programmatically, enabling scaled and controlled analysis without requiring CVD participants [3]. Sajek et al. use colour simulation models such as CIELAB to determine differences experienced under CVD conditions, providing physiological results rather than simple contrast ratio calculations [4]. Troiano et al. take a different approach, using generative methods to produce accessible colour palettes for users with CVD [5]. However, these studies focus on complete colour schemes rather than evaluating how specific interface components perform under simulated CVD conditions [4], [5]. This implies that while the perceptual and technical mechanisms of CVD are well known, their impact across specific UI components has not been empirically examined.

D. Research Gap

1) Evaluation Methods vs Component-Level Analysis: Taken together, these studies reveal a shared gap. Accessibility evaluation research has introduced clear methods and metrics, but these mainly assess entire webpages and do not focus on individual UI component performance [6], [2].

2) CVD Simulation vs UI Component Analysis: Reliable colour simulation methods have been developed from CVD research, but are usually applied to global colour schemes rather than specific UI elements [3], [4]. Therefore, neither approach explains how contrast changes across different component types or which components are most likely to fail accessibility requirements under CVD simulation.

This study addresses both gaps by combining CVD simulation with component-level contrast analysis across multiple websites [8], providing clearer guidance for accessible design. Fig2 illustrates these relationships and highlights the gap addressed in this study.

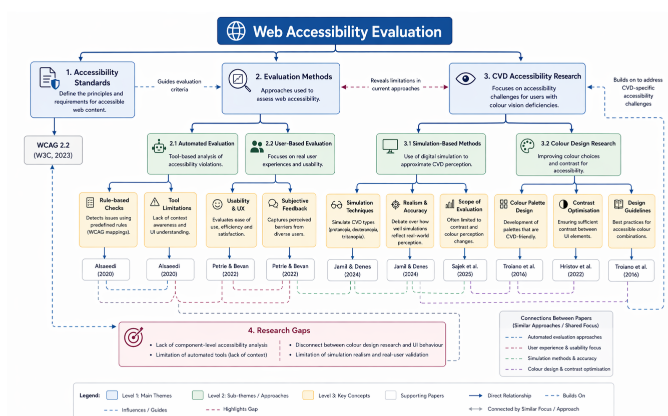

Fig. 2: Literature Map and Research Gaps

III. Research Methodology

A. Approach and Research Aims

This study aims to evaluate how different UI component categories perform in terms of foreground–background contrast under simulated CVD conditions [3], guided by the following questions:

- To what extent do UI components maintain acceptable contrast thresholds under simulated CVD conditions?
- How does contrast performance vary across component categories under simulated CVD conditions?

- Which UI components demonstrate the highest levels of contrast vulnerability across multiple websites?

A pragmatic, quantitative, and deductive methodology was used [2]. Accessibility was measured using contrast ratios, pass/fail outcomes, and WCAG 2.2 contrast-based success criteria, specifically (SC 1.4.3) Contrast (Minimum) and (SC 1.4.11) Non-text Contrast thresholds [1]. Rather than proposing new evaluation frameworks, this study applied WCAG criteria under simulated CVD conditions, building on [3] and [4]'s approaches.

B. Experimental Design

A dataset of 24 publicly accessible websites from several sectors including informational, commercial, and service-based domains. Websites requiring authentication and cookie-acceptance forms were excluded to ensure scanning consistency and completeness without interruptions.

To ensure domain diversity and reflect real-world variation in interface design [6], the dataset included websites from multiple categories:

- 3 – E-commerce (Amazon, Louis Vuitton, Printables)
- 2 – Travel and Booking (Ryanair, Emirates)
- 3 – News and Media (CNN, Times of Malta, Fox News)
- 4 – Educational and Informational (Wikipedia, Education.com, NASA, MDN)
- 5 – Developer and Technology Platforms (W3Schools, Firebase, Netlify, Stripe, Apple)
- 3 – AI and Software Services (OpenAI, Claude, Reddit)
- 1 – Government (MITA)
- 3 – Content Sharing and Creative Platforms (Behance, Imgur, Reddit)

An automated Python-based pipeline together with Playwright was used to scan and extract component styles once the whole website loaded, ensuring that slow-loading and dynamically generated content was captured. This approach enabled scalable and consistent evaluation, addressing limitations in manual and audit-based accessibility methods [6], [2].

Dichromacy models such as DaltonLens [9] were used to simulate extracted foreground-background element colours for 3 distinct CVD types (protanopia, deuteranopia and tritanopia) [3].

Initially each component's contrast ratio was calculated under normal, then simulated conditions and evaluated against WCAG thresholds (4.5:1 for standard text and 3:1 for large text). Contrast loss values were obtained, and components were classified into predefined categories (e.g., alert, button, card, form, input, link, navigation, text). Results were drafted into JSON then formatted into CSV format for statistical analysis in SPSS.

The pipeline consisted of the following stages:

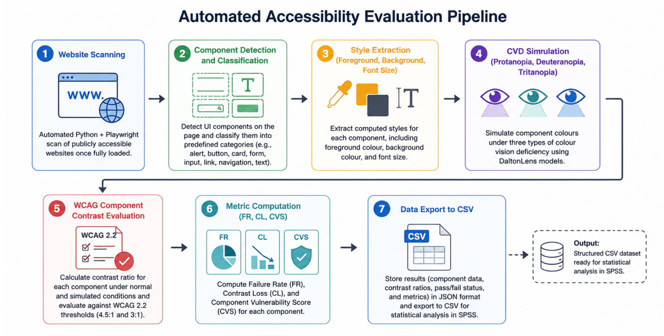

Fig. 3: Methodological Pipeline

C. Metrics

The primary metric was the WCAG Contrast Ratio (CR):

$$\text{CR} = \frac{L_{lighter} + 0.05}{L_{darker} + 0.05}$$

This metric measures the difference between foreground and background colours [1]. Higher values indicate better readability. It was selected as it allows results to be directly compared WCAG 2.2 contrast compliance standards. This metric was calculated for all 4 colour scenarios; normal, protanopia, deuteranopia and tritanopia and stored for further use in other formulae.

Three derived metrics were used to address the research questions:

a) Failure Rate (FR):

$$FR = \frac{\text{Failed Components}}{\text{Total Components}} \tag{1}$$

FR represents the percentage of components that do not meet WCAG contrast requirements. The results yield a measure of contrast performance and allows comparison between normal vision and simulated CVD conditions (RQ1).

b) Contrast Loss (CL):

$$CL = CR_{normal} - CR_{CVD} \tag{2}$$

CL measures how much contrast is lost under simulated CVD conditions compared to normal vision. Higher values indicate greater loss of visual clarity. This metric was used to compare the impact of different CVD types on contrast perception (RQ3).

c) Component Vulnerability Score (CVS):

$$CVS = \frac{\text{Components Failing under CVD}}{\text{Total Components of that Type}} \tag{3}$$

This metric indicates how vulnerable each UI component type is to contrast failure under CVD conditions. It enables comparison across component types and helps identify which elements are most at risk (RQ2).

Together, these metrics provide a structured evaluation of accessibility at a component level.

D. Method of Analysis

Statistical analysis was conducted using IBM SPSS Statistics tool [6]. Descriptive statistics summarised mean contrast ratios, standard deviations, and pass rates across CVD conditions and component categories.

A one-way ANOVA (Analysis of Variance) was conducted to determine whether contrast performance differed significantly across UI component categories, using contrast ratio and pass rate as dependent variables. The ANOVA produces an F-value, which represents the ratio of variance between group means to the variance within groups; higher F-values indicate greater differences between component categories. Statistical significance was evaluated using p-values, with a threshold of p¡.05, indicating that the observed differences are unlikely to have occurred by chance.

Where statistically significant results were identified, post-hoc tests were performed to determine which component categories differed from one another. To address RQ3, CL values were compared across the three CVD types (protanopia, deuteranopia, and tritanopia). Additionally, CVS was used to rank UI component categories based on how prone they are to contrast degradation to answer RQ2.

Results were reported using descriptive and inferential statistics, including means, standard deviations, F-values, and corresponding p-values to support interpretation of statistical significance.

E. Validity, Reliability, and Generalisability

Internal validity was supported through the use of established accessibility standards and models, including WCAG 2.2 contrast thresholds and validated CVD simulation techniques [1], [3], [4]. These result in contrast measurements that were based on recognised and widely accepted methods. However, the use of simulation introduces limitations, as it may not fully reflect the perceptual experiences and variability of real users with colour vision deficiencies.

Reliability was ensured through the fully automated evaluation pipeline. The same set of websites was analysed multiple times, with 5 repeated runs producing identical results. This consistency demonstrates that the method is stable and reproducible under the same conditions.

Generalisability was supported by the inclusion of a diverse dataset of websites across multiple domains, suggesting that the approach can be applied to different types of web interfaces. However, findings are limited to static visual contrast and do not account for dynamic interactions or other aspects of accessibility beyond colour contrast [1].

F. Ethical Considerations

No personal data or human participants were involved. All analysed websites were publicly accessible, and data collection adhered to ethical web usage practices.

IV. Findings and Results

A. Overview of Dataset

The dataset contained 22,950 evaluated UI components across three simulated CVD conditions: Protanopia, Deuteranopia, and Tritanopia. Components were classified into ten categories: *alert*, *button*, *card*, *form*, *header*, *input*, *link*, *navigation*, *other*, and *text*. The *other* category includes components (images) that did not clearly fit into any of the predefined categories. Each component was assessed against WCAG 2.2 minimum contrast thresholds [1].

B. Descriptive Statistics

Table I presents the mean contrast ratios, standard deviations, and pass rates across all three CVD conditions for the full dataset of 22,950 components.

TABLE I: Descriptive Statistics by CVD Condition (N = 22,950)

Condition	**Mean CR**	**SD**	**Min**	**Max**
Protanopia	11.14	6.03	1.00	20.82
Tritanopia	11.06	6.09	1.00	20.82
Deuteranopia	11.03	6.12	1.00	20.82
Pass rate: 85% across all three conditions (SD ≈ 0.354–0.359)				

Mean contrast ratios differed by less than 0.12, and pass rates were identical at 85% across all CVD conditions. The large standard deviations ($SD \approx 6.03$–6.12) relative to the means indicate high within-condition variability, suggesting that accessibility performance was affected more by component design decisions than by CVD type. These results suggest that simulated CVD types did not alter contrast performance significantly, which partially contradicts the original hypothesis that CVD conditions would reveal measurable differences across conditions. While overall accessibility levels were high, component-level differences in contrast performance were still observed.

C. Contrast Performance Across CVD Conditions

As illustrated in Fig 4, mean contrast ratios were near identical across all three CVD types for each component category. CL values consequently approached zero across the dataset. This finding indicates that dichromacy simulation models altered perceived hue without significantly affecting WCAG contrast ratios, aligning with Jamil's simulation-based observations [3]. This also supports Sajek et al.'s argument that WCAG's relative luminance formula only captures one dimension of perceptual colour difference [4].

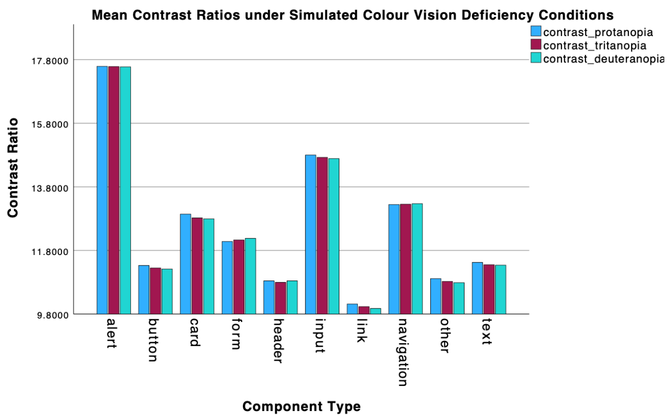

Fig. 4: Mean Contrast Ratios by Component Type under Simulated CVD Conditions.

D. Component-Level Analysis

Despite stable results, component-level analysis revealed meaningful variation in contrast performance across categories. Fig 5 illustrates pass rates per component type, and Table II presents the corresponding CVS derived from the SPSS Tukey HSD homogeneous subsets.

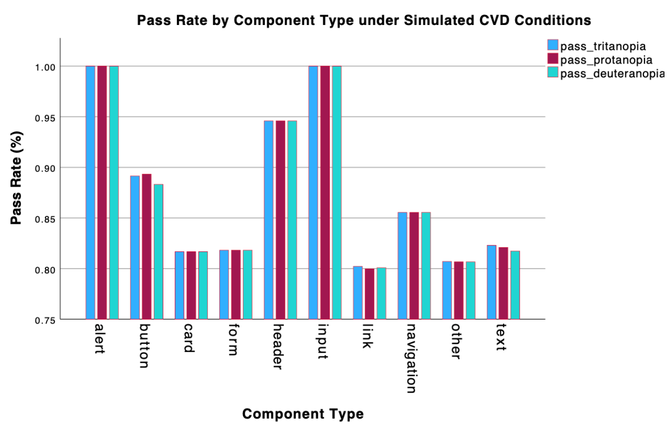

Fig. 5: Pass Rate by Component Type under Simulated CVD Conditions.

TABLE II: Component Vulnerability Scores (CVS) by Category — Protanopia Condition

Component	Mean CR (Protanopia)	Pass Rate (%)
Alert	17.59	100
Input	14.80	100
Navigation	13.24	86
Card	12.94	82
Form	12.08	82
Text	11.43	82
Button	11.33	89
Other	10.92	81
Header	10.85	95
Link	10.12	80

Alert ($M = 17.59$) and *input* ($M = 14.80$) components achieved 100% pass rates across all three CVD conditions, indicating that these functionally critical elements consistently applied sufficient contrast. This findings are consistent with Petrie and Bevan that interactive elements tend to receive greater design attention [2]. On the other hand, *link* components recorded the lowest mean contrast ratio ($M = 10.12$) and the lowest pass rate (80%), placing them in a lower-performing homogeneous subset in the Tukey HSD output alongside *other* and *header*. *Button*, *text*, *card*, *form*, and *navigation* occupied a mid-range band of 81–89%. The ranking was mostly identical the two other CVD types coming in at ($M_{link} = 9.98$) for Deuteranopia and ($M_{link} = 10.03$) for Tritanopia, confirming that the component hierarchy was not an artefact of one simulation condition.

In addition to this, *link* components recorded the highest CVS, indicating lower contrast performance compared to other component categories. This aligns with the vulnerability patterns identified in a study, which revealed that certain element types were more prone to WCAG failures than the aggregate page-level scores indicated [6].

E. ANOVA Results and Effect Size

One-way ANOVAs confirmed statistically significant differences in contrast performance across component types for all six dependent variables. Table III summarises the key results.

TABLE III: One-Way ANOVA Summary by Dependent Variable

Variable	F	df	Sig.	η^2
Contrast (Protanopia)	16.232	9, 22940	$< .001$	.006
Contrast (Deuteranopia)	16.637	9, 22940	$< .001$	.006
Contrast (Tritanopia)	16.219	9, 22940	$< .001$	.006
Pass Rate (Protanopia)	37.527	9, 22940	$< .001$	.015
Pass Rate (Deuteranopia)	28.645	9, 22940	$< .001$	.011
Pass Rate (Tritanopia)	34.927	9, 22940	$< .001$	.014

Note: In Table III, N represents the total number of UI components analysed. The F-value indicates the ratio of variance between component categories to variance within categories, where higher values suggest greater differences between component types. The significance value (p) represents the probability that the observed differences occurred by chance, with lower values (e.g., $p < .05$) indicating statistically significant results. The effect size (η^2) indicates the proportion of variance in contrast performance explained by component type, where higher values reflect a stronger effect and lower values indicate a weaker effect.

All six ANOVAs returned $p < .001$, confirming that component type significantly predicted both contrast ratio and pass rate outcomes. However, effect sizes were consistently small: $\eta^2 = .006$ for contrast ratio variables and $\eta^2 = .011$–$.015$ for pass rate variables. Under standard benchmarks, values below .01 are negligible and values between .01 and .06 are small. Post-hoc Tukey HSD testing identified link as significantly lower than button ($p < .001$), card ($p < .001$), navigation ($p < .001$), other ($p < .001$), and text ($p < .001$) across all three contrast ratio ANOVAs, placing it consistently in a distinct lower-performing homogeneous subset (Subset 1).

Alert was isolated in Subset 2, reflecting its significantly higher mean contrast ratio relative to the majority of other categories.

The small η^2 values raise an important internal validity concern. While component type was a statistically reliable predictor, it accounted for less than 2% of total variance in contrast outcomes. Uncontrolled design-level factors — including colour palette choices, CSS inheritance patterns, and front-end framework defaults — likely explained a substantially larger proportion of variance and were not isolated by this methodology [2]. This limits the degree to which component category alone can be used as a predictive accessibility risk factor.

F. Discussion in Relation to Hypothesis and Prior Studies

The original hypothesis that simulated CVD conditions would reveal measurable contrast performance differences across UI component categories, exposing patterns of accessibility vulnerability was partially supported. Component-level vulnerability patterns were confirmed and statistically significant. Link and other components were noticeably vulnerable relative to alert and input. However, the hypothesis that these patterns would differ across CVD conditions was not supported: contrast ratios and pass rates were stable across Protanopia, Deuteranopia, and Tritanopia, with no specific type producing significantly different outcomes.

This aligns with Jamil's findings [3] that dichromacy models primarily alter hue rather than luminance, and with Sajek et al.'s argument [4], WCAG contrast ratios do not fully capture CVD-relevant perceptual differences. On the other hand, Troiano et al. [5], worked at the palette level, and Alsaeedi's page-level metrics [6], the present study demonstrated that even within globally compliant colour schemes, component-specific failure rates varied meaningfully — a finding neither of those studies produced.

In terms of generalisability, the 22,950-component dataset spanning informational, commercial, and service-based domains strengthens external validity relative to studies using single sites or small datasets.

Overall, these findings demonstrate that while simulated CVD conditions do not significantly alter WCAG-based contrast outcomes, meaningful differences emerge at the component level, highlighting specific areas of accessibility vulnerability. This reinforces the importance of moving beyond aggregate or page-level evaluations and adopting more granular, component-focused analysis approaches. By identifying consistent patterns of underperformance across UI components, this study contributes to a more precise understanding of how accessibility issues manifest in practice, providing a foundation for more targeted design and evaluation strategies in future research and development.

V. Conclusion

This study aimed to evaluate how different UI component categories perform in terms of foreground-background contrast under simulated CVD conditions, using WCAG 2.2 standards as the benchmark [1]. Based on the analysis of 22,950 interface components across 24 real-world websites, several key conclusions can be drawn.

The primary conclusion is that overall contrast performance remains stable across simulated CVD conditions. Mean contrast ratios and pass rates (85%) barely differed across Protanopia, Deuteranopia, and Tritanopia. This suggests that WCAG-compliant designs generally maintain acceptable levels of contrast regardless of the simulated type of colour vision deficiency. At an aggregate level, accessibility performance appears consistent and relatively high.

However, this consistency does not extend to the component level. The findings demonstrate that accessibility is not uniformly distributed across UI components. Instead, performance varies significantly depending on the component category. In particular, *alert* and *input* components achieved consistently high contrast performance, reflecting strong design prioritisation for critical and interactive elements. In contrast, *link* components were identified as the most vulnerable, consistently recording the lowest contrast ratios and pass rates across all simulated conditions. Other components such as *header* and *other* also demonstrated comparatively lower performance, while *button*, *text*, *card*, *form*, and *navigation* occupied a moderate performance range.

In relation to the research questions, the findings provide clear answers. For RQ1, the majority of UI components maintained acceptable contrast thresholds under simulated CVD conditions, as evidenced by the overall 85% pass rate, although some failures persisted. For RQ2, contrast performance varied significantly across component categories, with statistical analysis confirming that component type influenced contrast outcomes. For RQ3, *link* components demonstrated the highest level of contrast vulnerability, consistently underperforming across all websites and conditions.

The original hypothesis was partially supported. The study successfully identified measurable differences in contrast performance across UI component categories, confirming the presence of component-level accessibility vulnerabilities. However, the expectation that contrast performance would differ across simulated CVD conditions was not supported. The results showed minimal variation between Protanopia, Deuteranopia, and Tritanopia, indicating that luminance-based contrast metrics, as defined by WCAG 2.2, are not sensitive to the perceptual differences introduced by dichromatic vision simulations.

Despite these findings, several methodological limitations should be acknowledged. First, the study relied exclusively on WCAG luminance-based contrast ratios, which do not fully capture perceptual differences caused by CVD. As a result, the simulation of different CVD conditions had limited impact on the measured outcomes. Second, the analysis was restricted to statically rendered components and did not include dynamic states such as hover effects, focus indicators, or animations, which may influence accessibility in real-world interaction. Third, unequal sample sizes across component categories may have influenced statistical comparisons, particularly for categories with very small representation. Finally, design-specific factors such as colour palette choices and CSS inheritance were not controlled, meaning that a significant portion of variance in contrast performance remains unexplained.

Future research should explore more perceptually accurate evaluation methods, such as colour difference models (e.g., CIELAB or ΔE), which may better capture accessibility issues experienced by users with CVD. Additionally, incorporating dynamic interface states and conducting user-based testing with individuals who have diagnosed CVD would provide a more comprehensive and ecologically valid assessment of accessibility. Further investigation into adaptive accessibility solutions, such as real-time colour adjustment tools or browser extensions, could also offer practical approaches to addressing the limitations identified in this study.

In conclusion, while WCAG 2.2 contrast guidelines appear effective in ensuring baseline accessibility, this study demonstrates that important component-level vulnerabilities remain. Furthermore, the limited sensitivity of luminance-based metrics under CVD simulation highlights the need for more comprehensive evaluation approaches to support inclusive design.

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